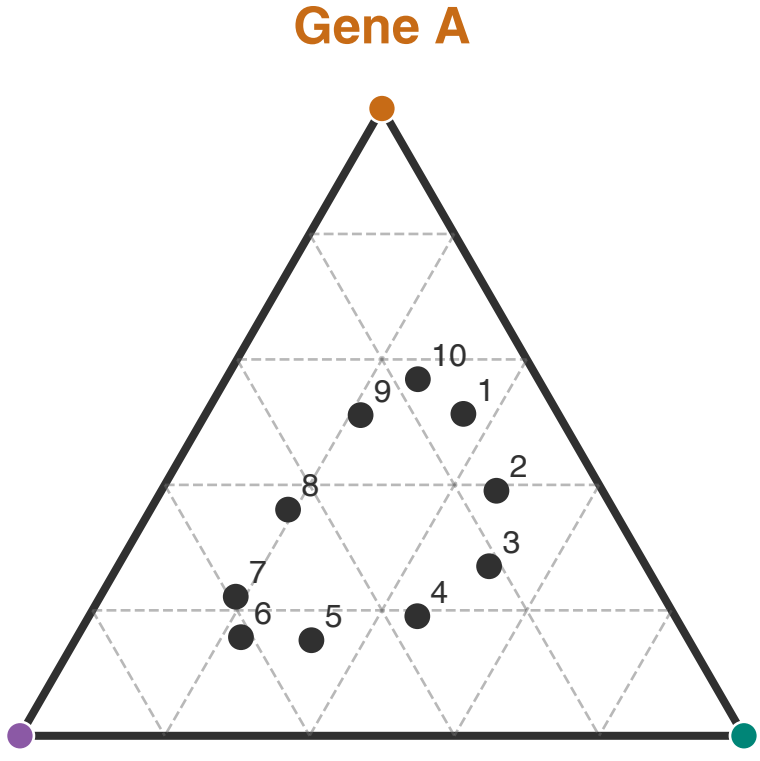
1. Gene expression in CLR space



$$z_{ig} = \log \frac{x_{ig}}{g(\mathbf{x}_i)}$$

$$g(\mathbf{x}_i) = \left(\prod_g x_{ig}\right)^{1/G}$$